Part: **4.9 – COMPOSITE PARTS**

Chapter: **4910**

Title: **FATIGUE AND DAMAGE TOLERANCE OF COMPOSITE PARTS. INTRODUCTION.**

Revision: **1.0**

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1 PREAMBLE

This chapter presents the methodology to be considered for the fatigue assessment of composite structures.

The fatigue, understood as the progressive deterioration of the material strength due to repetitive loads, is a concept applicable to both metallic and composite materials. The certification requirements intends to achieve the same level of reliability for composite structures than the required for metallic ones, but the accurate prediction of the fatigue behaviour of composites becomes challenging because of the complex failure mechanism, the material properties sensitivity to environmental conditions, such as temperature and moisture, and the scatter in the material fatigue data.

According to AMC 20-29 Section 8: *“Composite damage tolerance and fatigue evaluations require substantiation in component test unless experience with similar design, material systems and loading is available to demonstrate the adequacy of the analysis supported by coupons and sub-component tests”.*

Applicable requirements, associated means of compliance and justification strategies shall be different for different structural parts and details depending on several aspects, including: specific materials, production processes, inspection procedures, and structural significance of the part in consideration.

2 INTRODUCTION TO CERTIFICATION ASPECTS

This section does not intend to discuss exhaustively the considerations related to structure certification/qualification, as each programme shall deal with these topics in their respective Certification/Qualification Summary Plan (out of the scope of this document). General advice can be found in airworthiness regulations (Ref. 1, Ref. 2…), further guidance can be found in justification strategies and methods for F&DT justification of composites in CMH-17 (Ref. 5 , in particular Vol. 3, chapter 12)... It has been considered useful, however, to summarize here some key points which shall typically affect the fatigue and damage tolerance aspects to cover within the part of the structural substantiation made by "conventional analysis", identified as "deterministic no-growth approach" in CMH-17 §12.2.4.1 (the part covered by probabilistic analysis and by advanced numerical simulations are out of the scope of this document).

2.1 STRUCTURAL CLASSIFICATION

The following definitions summarize a categorization of the structural components in use within Airbus for certain programmes (*), extending the traditional separation in primary and secondary structures defined in Certification Specifications (Ref. 3):

- **Primary structures (or category A)**: Structures identified as Principal Structural Elements (PSE). These are elements that contribute significantly to carrying flight, ground or pressurization loads and whose failure could result in catastrophic failure of the aircraft.

- **Secondary structures**: those out of the scope of the definition above. They are typically sub-classified within Airbus, according to one of the following categories:
  - ✓ Category B: Structures whose failure or detachment could indirectly compromise continued safe flight or landing by an adverse effect on a Category A structure
  - ✓ Category C: Structures whose failure or detachment will not compromise continued safe flight or landing but where the potentially large size of released elements needs to be considered. As these structures are not identified as either category A or B, any failure or departure from the aircraft must be demonstrated as not preventing continued safe flight and landing and the probability of occurrence is acceptably low. No detachment of structure is allowed
  - ✓ Category D: Structures whose failure or detachment has no airworthiness consequences but only has an economic impact.

Remarks and footnotes:

- The correspondences done above between EASA/FAA (Primary/Secondary) and Airbus (*A/B/C/D*) classifications are just indicative. There are no formal equivalences among them.
- Damage tolerance justification is not required for parts classified as *C* (**) or *D*
- Fatigue justification (damage initiation) is not required for parts classified as *D*

(*) The extended classification (categories *A, B, C* and *D*) has not been applied to legacy combat aircrafts within Airbus, being on the other hand a consolidated standard in commercial aircrafts

(**) This relaxation is not applicable to the means of attachment of a class *C* structure to the principal structure

2.2 DAMAGES CLASSIFICATION

The fatigue and damage tolerance evaluation of a composite structure shall begin with a damage threat assessment to identify all the possible damages that could appear during the manufacturing, assembly and in service. See a more detailed discussion on this topic in chapter 4920 “COMPOSITE DAMAGE THREATS”

The damages are classified into five categories, depending on their detectability thresholds. Each category has dedicated structural requirements to be demonstrated to be in compliance with applicable regulations (AMC 20-29, Ref. 2). Further discussion and details about damage types and associated approaches for structural justification can be found in chapter 4930 “DAMAGE TOLERANCE JUSTIFICATION OF COMPOSITE PARTS”.

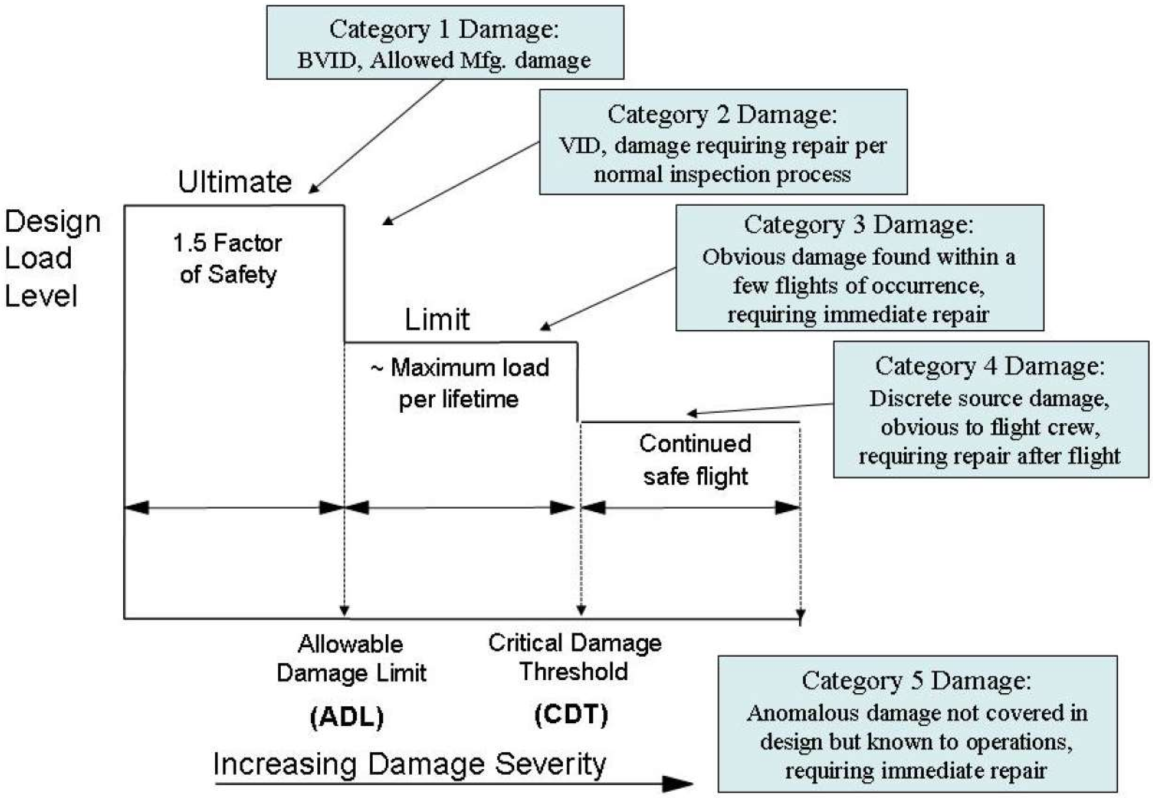

Figure 1. Damage classification attending its severity

Notice that all damage that lowers the strength below ultimate load (i.e. damages from Category 2 and beyond) must be repaired when found.

2.3 THE DAMAGE-NO-GROWTH APPROACH

The slow/stable growth of damages is a behaviour not commonly observed for the composite materials typically used in the aerospace industry. The prediction of the operation time taken by an undetected damage to become of a size detectable is principally, today, in the field of research. Thus, as noted by these extracts from Ref. 5 / CMH-17 - Vol. 3…

§12.1.3 *"In the absence of predictive tools for growth, design values are typically established with sufficient margins to ensure that damage growth due to repeated loads will not occur. This design/certification approach for avoiding the potential growth of damage is known as the "no-growth" approach. It has been practical for most composite designs, which have proved to be fatigue insensitive at typical design stress levels."*

§12.3.2.1/2 *"Repeated load capability associated with … damage is usually demonstrated through component and full-scale testing at room-temperature".*

§12.6.4.2.2 *"Using threshold values established at the coupon level to make analytical predictions demonstrating "no growth" at a structural level may be an effective way to increase confidence in a single full scale test used for certification."*

In best cases, material curves for addressing the fatigue life (no-damage-initiation) and/or the structural damage tolerance (no-damage growth) may be acceptable and life ratios can be calculated. The fatigue allowable values historically used in Airbus DS to certificate the use of composite structures are based on the no-growth approach. The no-growth demonstration should be based on coupon, element or sub-component level tests, or based on previous experience in case of similar design.

Residual strength associated to Category 1 damages (BVID and manufacturing anomalies) is demonstrated by analysis using design values obtained in coupons and element tests (again, analytical predictions of the residual strength of structures including low velocity impact damages is principally, today, in the field of research). Structural capability with damages of greater categories are generally addressed using validated analysis methods and/or directly by test.

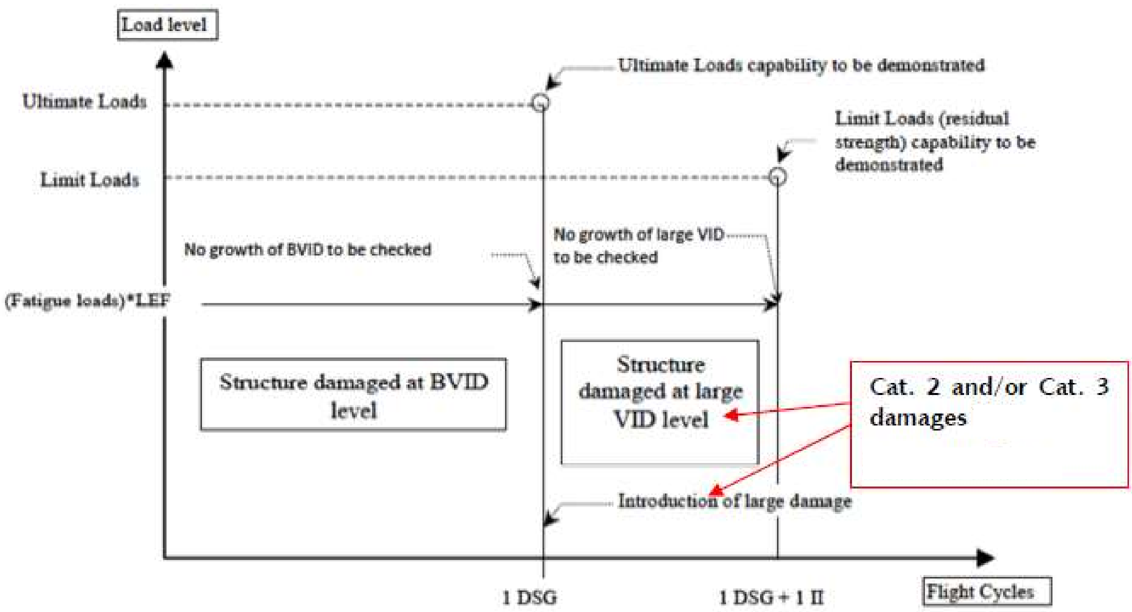

Figure 2. Damage Tolerance philosophy for composite structural parts

3 INTRODUCTION TO F&DT ANALYSIS IN COMPOSITE MATERIALS

Structural justifications of composite parts shall include static strength and residual strength analysis of parts with allowable quality/manufacturing defects, and also the consideration of accidental damages of different kinds. Advice of stress methods and material design values can be found in Volumes 1 and of this Manual. Material response under cycling loads shall have to be addressed also as a part of the structural justification (this part the more commonly known as Fatigue and Damage Tolerance justification), and will not necessarily be done exclusively by testing.

Fatigue analysis methodologies, and associated design values have not the same degree of maturity as their alternatives in the scope of static strength.

✓ **Fibrous composite laminates**

When talking about composite materials, most of the times the phrase refers to fibrous composite materials, i.e. to materials composed of reinforcement fibres (typically of carbon or glass, sometimes arranged as woven fabrics) embedded in matrix of a different material (typically polymeric resins), arranged as solid laminates.

Good design practices and structural sizing minimize interlaminar stresses within these materials. As for the static strength justification, the relevant internal load to use in fatigue justification is the strain level within the laminate (axial strain along fibre direction in particular). Fatigue/no-growth allowables shall be typically given (or converted to if necessary for practical use) in terms of allowable strains (as a threshold), or S-N curves in best cases.

✓ **Bonded joints**

No accepted analysis method for fatigue justification of bonded joints is available (dedicated fatigue allowables are typically missing). Substantiation is, most of the times, supported exclusively by testing. In best cases, some simple practical design values, as stress thresholds, comparable to the mean/averaged shear stress in a well-design adhesive joint, would be available (see also Ref. 7).

Adhesive joints in damage tolerant structures, in contrast with cohesive joints (e.g. the interlaminar attachment between two plies of a laminate, including the interphase between cocured laminates), shall typically be designed to retain a residual strength capability up to Limit Load with the bonded joint assumed lost up to the existing damage arresting features (noted in AC 20-107B and adopted from US regulations of general aviation 14 CFR § 23.573 or its equivalent European norm Ref. 6).

✓ **Bolted joints**

No accepted analysis method for fatigue justification of bolted joints is available (dedicated fatigue allowables missing). F&DT substantiation is supported nearly exclusively by part/component testing. It is noted, however, that properly design bolted joints are quite insensitive to fatigue.

✓ **Sandwich panels**

The joint between the sandwich faces (typically fibrous composite laminates) and the core is treated as an adhesive joint, and the associated demonstration of residual strength assuming a massive disbond makes this configuration, most of the times, not compatible with damage-tolerant parts.

Strain levels at faces are used for the no-damage-initiation/growth demonstration when applies (see discussion on composite laminates above on this page).

A kind of rule-of-thumb which may practical for preliminary design phases, observed throughout tests programs performed on different types of specimens and composite materials within Airbus and recognized in literature (e.g. see CMH-17 Vol. 3 §12.6.3.7), is that it can be expected that a composite material shall not suffer damage initiation or growth with cycling loads up to one half of its static ultimate strength.

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
BVID		Barely Visible Impact Damage, i.e. visually detectable during maintenance, or manufacturing, programed inspections
DSG		Design Service Goal
FH	hrs	Flight Hour
LEF		Loading Enhancement Factor. In this scope, the factor applied to nominal fatigue loads in tests in order to cover material dispersion, this approach frequently used as alternative to prolonging the test duration by applying a scatter factor in the test cycles (what is usually done in fatigue tests with metallic parts).
LF		Life Factor
F&DT		Fatigue and Damage Tolerance
LL		Limit Loads
PSE		Principal Structural Element (per AC 25.571 certification standard), an element that contributes significantly to the carrying of flight, ground, or pressurization loads, and whose integrity is essential in maintaining the overall structural integrity of the airplane. Principal structural elements include all structure susceptible to fatigue cracking, which could contribute to a catastrophic failure.
S-N		Curve defining the material fatigue performance in terms of cyclic stress (*S*) against the cycles to failure (*N*), also known as Wöhler curve. Within this chapter it should be interpreted in a broader manner, with the *S* a magnitude representing a generic internal load (including stress but also, typically, strain), the most convenient for the design detail of interest
SSI		Structural Significant Item (in the airframe maintenance program), any detail, element or assembly, which contributes significantly to carrying flight, ground, pressure or control loads, and whose failure could affect the structural integrity necessary for the safety of the aircraft. SSIs must not be confused with Principal Structural Elements (PSE), however, all PSEs must be addressed by the SSIs.

UL	Ultimate Loads
VID	Visible Impact Damage

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	[EASA] CS-25	Amd.17	Certification Specifications CS-25 § 25.301, 25.302, 25.303, 25.305, 25.307 (static strength) and 25.571 (fatigue and damage tolerance)
Ref. 2	[EASA] AMC 20-29	2010	COMPOSITE AIRCRAFT STRUCTURE (or FAA Advisory Circular AC No: 20-107B)
Ref. 3	[Airbus] AM2396.1	Is. A	Structure Analysis Categorization for Definition, Justification and Certification Phases
Ref. 4	DOT/FAA/AR-10/6	2011	Determining the Fatigue Life of Composite Aircraft Structures Using Life and Load-Enhancement Factors
Ref. 5	CMH-17	2012	COMPOSITE MATERIALS HANDBOOK (VOLUME 3. POLYMER MATRIX COMPOSITES MATERIALS USAGE, DESIGN, AND ANALYSIS)
Ref. 6	[EASA] CS-23	Amdt.4	Certification Specifications and Acceptable Means of Compliance for Normal, Utility, Aerobatic, and Commuter Category Aeroplanes (inc. CS 23.573 “Damage tolerance and fatigue evaluation of structure”)
Ref. 7	TAE-T-MM-160005	Is. B	GUIDELINES FOR THE SIZING AND STRUCTURAL ANALYSIS OF BONDED JOINTS AND REPAIRS

SOFTWARE

Program	***Version***	***Description***

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
1.0 23/11/2021	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Analysis (TEPAA)	José Efraín Mirón Rubio 23/11/2021	Holger Hickethier 22/11/2021
Stress Methods (TEPAP-TL4)	Darío R. Fernández 22/11/2021	-

Table 3. *Approval signatures table for last revision of this chapter*